Code-Layout, Readability and Reusability

Date	10 July 2026
Team ID	xxxxxxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Code Layout Checklist

S.No	Code Quality Parameter	Followed	Remarks
1	Consistent Indentation	Yes	PEP8-style 4-space indentation used throughout train_model.py and app.py
2	Proper File Structure	Yes	Separated into model/ (training + deployment) and flask_app/ (web app) folders
3	Meaningful Variable Names	Yes	e.g. X_train_bal, fraud_probability, best_model_name
4	Function / Method Names	Yes	load_data(), preprocess(), train_and_compare(), prepare_row()
5	Code Comments	Yes	Docstrings on every script plus inline comments explaining SMOTE, scaling, thresholding
6	Modular Design	Yes	Preprocessing, training, evaluation, and saving are separate reusable functions
7	No Redundant Code	Yes	Single shared preprocessing function used by both training and Flask inference
8	Error Handling	Partial	Flask /predict route wraps input parsing in try/except; batch route can be hardened further

Reusable Components / Modules

S.No	Component / Module	Language/Tech	Reusability	Where Reused
1	preprocess()	Python/pandas	High	Used in both train_model.py and app.py's prepare_row()
2	StandardScaler (scaler.pkl)	scikit-learn	High	Reused at inference time to match training-time scaling
3	Model comparison loop	Python	Medium	Reusable pattern for adding new algorithms in future

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clear separation of training, deployment, and web app code
Readability	5	Descriptive names, docstrings, consistent style

Reusability	4	Preprocessing logic shared between training and serving
Documentation / Comments	5	Every script has a module docstring and inline explanations
Overall Score	4.75	